

TITLE: Wind-Induced Pressure Analysis and Structural Performance Evaluation of a Multi-storey Building

MADHU NANDAN M | MOHAMMAD YASEEN D G | THARUN N | SUPHIAN J

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Introduction

This project focuses on how wind forces affect tall/mid-rise buildings, specifically a **G+9** residential building. Wind is one of the most critical lateral loads that act on multi-storey structures,

IS 875 Part 3 provides standard wind load calculations.

CFD can predict actual wind behavior around buildings



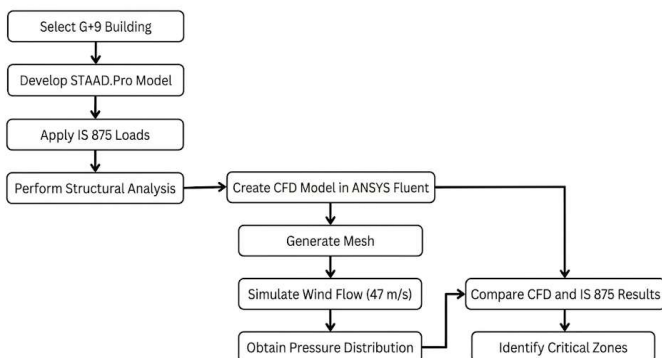
Problem Definition

- **IS 875** uses generalized **pressure coefficients (Cp)**.
- Actual **building geometry** affects wind pressure distribution.
- **Corner vortices** and **flow separation** are often neglected.
- This may result in **inaccurate structural design**.

Objectives

- Perform Structural analysis of **G+9 residential building** in **STAAD.Pro** considering it is in zone 4
- Perform **CFD analysis** using **ANSYS Fluent**.
- Extract **pressure coefficients (Cp)**.
- Compare **CFD results** with **IS 875** values.
- Compare Results with different Architectural Design

Methodology



STRUCTURAL ANALYSIS:

1. Building Selection
2. Structural Modeling
3. Load Application
Applied Dead Load, Live Load, and Wind Load as per IS 875.
4. Structural Analysis
Evaluated Displacement, Drift, Bending Moment, and Shear Force.

CFD:

1. **CFD Model Development**
Created building geometry in ANSYS Fluent.
2. **Mesh Generation**
Generated computational mesh.
3. **Wind Simulation**
Applied 47 m/s wind speed (Wind Zone 4).
4. **Pressure Evaluation**
Obtained Pressure Distribution and Pressure Coefficients.
5. **Comparison & Retrofit**
Compared CFD and IS 875 results.
Proposed suitable Retrofitting Measures.

Tools used

AUTO CAD - For Drafting and Planning.



STAAD.Pro – For Structural Analysis.

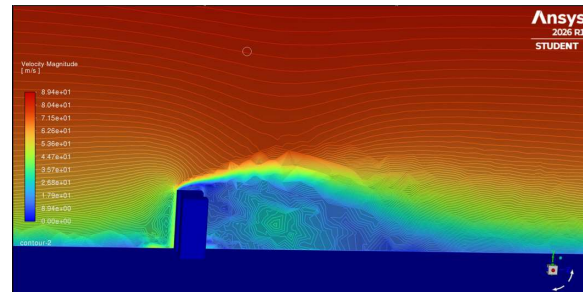


ANSYS Fluent – For Computational fluid Dynamics analysis.



Results and Discussions

- Maximum pressure on the windward face and suction on the leeward face.
- High pressure observed at corners and edges.
- CFD results differed from IS 875 due to actual airflow effects.
- Maximum displacement occurred at upper storeys



Conclusions

- CFD analysis helped visualize the **pressure distribution, velocity flow, and wind effects** around the building.
- Comparison between **CFD-derived pressures** and **IS 875 values** highlighted the influence of actual airflow behavior on structural loading.
- Structural analysis identified critical parameters such as **maximum displacement, support reactions, bending moments, and storey drift**.
- The study demonstrated that integrating **CFD analysis** with **structural analysis** provides a more realistic assessment of building performance under wind loading.
- The proposed methodology can be used for improving the safety and reliability of multistorey buildings.

Outcome of the work

- Accurate assessment of wind pressure distribution on the G+9 building using CFD.
- Identification of critical zones prone to high pressure, suction, and vortex formation.
- Evaluation of building displacement, drift, and support reactions under wind loading.
- Validation and comparison of CFD results with IS 875 wind load provisions.
- Better understanding of wind-structure interaction for safer multi-storey building design.

References

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